

MyDesign Portfolio

Element K: Reflection on the Design Project

Engineering Design Process Summary (K1, K2)

- Presenting & Justifying a Problem & Solution Requirements (elements A-C)

We started by defining the problem at hand and determining the importance of the problem. The problem was easy to define as the mic flags that Mr O'Farrell had were clearly scratched and peeling. The past flags also were able to give us ideas for requirements to put in our problem statement such as size and weight requirements. Between the clear problems and strengths with the past design we made our problem statement which was to design a durable, lightweight, replaceable, secure mic flag that communicates and promotes the TimberTV and Ambush News' logo and allows broadcasters to have multiple flag options to choose from when recording with quantifiable constraints such as weighing under 100 grams and having a panel for the logo that is at least 75mm by 50mm. Finding the impact of mic flags was a lot tougher to do. There is very little research on mic flags as they are a simple product with really only one use. Through our research, we were able to define the main, and only true, impact of a mic flag is to advertise the logo on them. After creating a problem statement and finding the impact of mic flags, we looked into past mic flag designs to assess the strengths and weaknesses of them. We looked at five prior design attempts and listed what the strengths and weaknesses for each one were. We found out different ways to secure the flags such as metal clamps, rubber arms, and foam. We also looked at different ways to design a mic flag such as removable logos and secure logos. This step was helpful in the process as we were able to use the good ideas from these steps for our mic flag while also removing failures of past designs. After looking at past designs, we listed out different requirements for our design. This step was a bit difficult as we struggled to find 7 different requirements for our flag. This led to some weaker design requirements, such as the corners must not be sharp, being added to fulfill the number of requirements needed. This step was very helpful however as we were able to figure out with Mr O'Farrell what he wanted out of the flag. Using the ranking of the requirements from Mr O'Farrell, we knew that when we were designing our flag Mr O'Farrell would be satisfied with the final result as he had given us which requirements he wanted the most for the mic flags.

- Design Concept Generation, Analysis, and Selection (element D)

This part of the project was easily the most time consuming. We did a good job creating many ideas which gave us a lot of room to improve our project. Since we had several ideas from the start, we were able to combine elements from the ideas to make the best idea possible. The selection of the idea was quite easy as we all felt like the sliding panels would be the best idea and the easiest in practice. We also chose to look further into the LED one because we had a lot of faith that it would look really good even if it wasn't super practical. We are glad that we kept the idea as Mr O'Farrell gave us a suggestion to make it multi-colored with two separate prints for the inside and outside. I think our choice of selection grid where we are scoring the designs in different categories instead of ranking them was the better grid choice of the two. This allowed different flags to showcase where they thrived and struggled instead of just being ranked in each category. If we had ranked them, certain designs would have scored very differently as they would have been capped due to other designs being better or saved due to other designs being worse rather than a good design being scored well and a bad design being scored badly. Finding the price of each design was tough as we needed to find out the total price for each material and find out how much we used. While this is easy to do for 3D print filament and foam, it is tough to gauge how much money in tape or ink that we used. The plan of action was the final part of this section. The plan of action was easy to put together as we knew what we needed to do, but finding how much time each step would take was difficult as we had to estimate how long it would take us to do certain tasks that we had never done before.

- Constructing and testing prototype (elements G-I)

First we sketched the design of the mic flags on paper, we then made a cardboard version of mic flag we sketched on the paper, next we then saw how well mic flag would fit on a regular microphone and copied the design into onshape with the correct dimensions based on the cardboard flag, we then placed the 3d designs into assembly on onshape to confirm the dimensions, next we exported the base of the design and rendered the base in prusa to 3d print the design out, finally we 3d printed the design and took notes on what was wrong, if the size or width was off we would change those based on how the mic flag held up on the microphone. We then would test our mic flag in a couple of different ways, some being more of a priority than others. The first priority is to make the pictures on the mic flag not fade, we tested this by leaving one of the flags in a car for at least 24 hours to see how it would affect the flag. Second priority is for the size of the logo being 75mm by 50mm, we tested this by using a digital multimeter to find the dimensions of the panel to make sure it's big enough to fit the logo. Third is for the mic flags to be durable, we tested this by simply dropping the mic flag from a couple feet to see if anything chipped or broken apart.

Fourth, and our first moderate priority, is for the mic flags to be replaceable when damaged, we test this by starting a timer and seeing how long it takes for the 3d printer to print out a new mic flag, if it is under 24 hours then it is a success but if it is above then it fails. Fifth is for the mic flag to be able to fit on multiple different mic holders, we test this by putting the mic flag on a mic holder, and shaking it for a minute, we will test this with 3 different mic holders and see if it stays on firmly, if it does then it is a success. The 6th priority, and our first low priority, is for the mic flag to not be sharp on the ends so no one gets hurt by it. We tested to make sure it wasn't sharp by pressing the edges of the flag against cardboard to see if it would make a mark on said cardboard. The 7th and last priority is for the mass of the mic flag to be below 100g, we tested this by simply putting the flag on a digital scale to see if it met the requirements. Those are all the design requirements from biggest to smallest priority.

- Documentation of Evaluation (element J)

After presenting our project, we got feedback on our project. I wish we would have gotten this feedback earlier on as we would have had more time to fully implement the ideas given such as leaving the logos in a car windshield so we could test if the sunlight or heat would impact the logo. While it is nice to get feedback that is overwhelmingly positive, I wish we would have gotten more feedback on how we can improve our project. We did not get any critical feedback and did not get any ideas on how to improve outside the windshield idea. I wish we would have gotten more critical feedback so we could make our project improve further rather than staying stagnant after getting feedback.

Lessons Learned About the Engineering Design Process (K3)

Ideas begin with any possibilities no matter how crazy

It is important to hear and work with everyone involved with this project, like your group members, stakeholders, and field experts

It's important to be able to change ideas and adjust to outside aspects

The engineers should always be learning from their projects